YEAR 7 MATHS · CHAPTER 5 · MIND MAP

Decimals

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1 Place value and decimals

After the point: tenths, hundredths, thousandths.

3.47 is 3 ones, 4 tenths and 7 hundredths

0.5 is bigger than 0.45

Watch out: more digits doesn't mean bigger.

2 Rounding decimals

Look at the next digit: 5 or more rounds up.

3.476 to 1 decimal place is **3.5**

6.849 to 2 decimal places is **6.85**

2.93 to the nearest whole is **3**

3 Adding and subtracting

Line up the decimal points. Fill gaps with zeros.

$3.5 + 2.75 = 6.25$

$5 - 1.8 = 3.2$

4 × and ÷ by 10, 100, 1000

Multiply: digits move left.
Divide: digits move right.
One place for each zero.

$3.4 \times 100 = 340$

$56 \div 10 = 5.6$

$0.7 \div 100 = 0.007$

5 Multiplying decimals

Multiply as whole numbers, then count the decimal places.

$0.3 \times 0.4 = 0.12$ **(two places)**

$1.2 \times 3 = 3.6$

$2.5 \times 1.2 = 3$

6 Dividing decimals

Make the number you're dividing by whole: move both points the same way.

$7.5 \div 3 = 2.5$

$4.2 \div 0.7 = 42 \div 7 = 6$

7 Decimals and fractions

To turn a fraction into a decimal, divide the top by the bottom.

$0.75 = \frac{3}{4}$ $0.2 = \frac{1}{5}$

$\frac{3}{8} = 0.375$ $\frac{2}{3} = 0.666\ldots$

8 Decimals and percentages

Decimal to percentage: multiply by 100. Percentage to decimal: divide by 100.

$0.35 = 35\%$ $1.2 = 120\%$

$7\% = 0.07$

9 Expressing proportions

Write the part over the whole, then convert.

12 out of 20: $\frac{12}{20} = \frac{3}{5} = 0.6 = 60\%$

9 out of 25: $\frac{9}{25} = 0.36 = 36\%$